

OSCODA-WURTSMITH, MI, USA

WMO: 726395

Lat: 44.450N

Lon: 83.400W

Elev: 633

StdP: 14.36

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 14808

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.0	1.0	-10.8	3.1	-3.3	-6.3	4.0	3.9	25.9	24.1	23.9	24.1	6.8	240	0.587

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	89.6	71.9	85.8	70.3	82.2	68.6	75.0	84.8	73.1	81.9	71.2	78.8	13.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.2	122.0	80.2	70.3	114.2	77.8	68.6	107.7	75.9	38.8	85.4	37.0	82.3	35.4	79.1	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	24.1	20.5	18.4	DB	-11.2	94.2	6.6	3.6	-15.9	96.8	-19.8	98.9	-23.4	100.9	-28.2	103.5
(5)				WB	-11.7	77.8	6.5	2.4	-16.4	79.6	-20.1	81.0	-23.8	82.3	-28.5	84.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.5	21.2	22.4	30.6	41.7	53.5	63.5	69.0	67.5	60.8	48.8	37.7	27.3	
	DBStd	19.00	10.83	10.42	10.52	8.79	8.80	7.67	6.49	5.78	7.73	8.46	8.39	9.52	
	HDD50	3827	894	773	607	279	58	1	0	0	6	129	376	704	
	HDD65	7545	1359	1193	1065	699	373	120	33	38	172	507	818	1168	
	CDD50	2173	0	0	6	31	168	406	589	543	329	92	8	1	
	CDD65	418	0	0	0	1	18	75	157	116	46	5	0	0	
	CDH74	3961	0	0	4	32	260	806	1474	955	386	44	0	0	
	CDH80	1254	0	0	1	6	79	284	500	271	102	11	0	0	
Wind	WSAvg	8.2	9.2	9.2	8.7	9.2	8.0	6.9	6.8	6.7	7.1	8.4	8.8	9.0	
Precipitation	PrecAvg	30.2	1.8	1.5	1.7	2.7	3.1	3.3	3.3	3.1	2.8	2.8	2.4	1.8	
	PrecMax	36.9	3.0	3.2	4.9	4.7	6.6	7.0	6.4	5.9	5.1	6.4	5.3	4.0	
	PrecMin	24.8	0.2	0.4	0.5	1.1	1.3	0.9	1.1	1.2	0.4	0.8	0.7	0.4	
	PrecStd	3.1	0.9	0.7	1.1	0.9	1.3	1.4	1.2	1.1	1.2	1.4	1.3	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.2	53.6	69.7	79.2	88.0	91.3	93.3	91.1	88.3	79.4	65.3	55.2	
		MCWB	45.8	50.1	54.3	59.8	67.8	72.6	74.4	72.8	71.8	64.5	56.0	51.9	
	2%	DB	43.1	44.7	57.3	70.0	81.4	87.9	89.8	86.5	82.2	72.3	59.1	48.0	
		MCWB	39.6	39.8	47.8	55.7	64.1	70.6	72.8	72.1	69.0	62.1	52.1	44.8	
	5%	DB	38.4	39.3	51.8	63.1	75.1	82.4	85.6	82.2	78.8	67.3	54.6	43.4	
		MCWB	36.1	35.8	45.7	52.4	61.7	68.2	70.3	69.4	66.9	60.4	49.5	40.3	
	10%	DB	36.1	36.5	45.3	56.7	70.1	78.9	81.9	79.4	74.3	63.3	50.7	39.4	
		MCWB	34.4	34.0	39.7	47.5	59.0	65.8	69.0	67.8	65.1	57.0	46.7	36.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.4	49.7	58.9	63.3	70.9	75.5	78.3	76.4	74.4	69.8	58.3	53.5
			MCDB	47.0	54.5	66.1	73.7	81.9	87.8	89.3	85.0	83.9	75.8	63.2	55.4
2%		WB	40.2	40.2	51.0	58.0	67.5	73.1	75.7	74.2	71.8	64.6	53.7	46.0	
		MCDB	42.7	43.9	57.2	66.7	76.0	83.9	85.2	83.2	79.1	68.9	58.0	47.5	
5%		WB	36.6	36.8	45.3	53.1	64.2	70.6	73.3	71.9	69.6	61.1	50.0	41.0	
		MCDB	38.3	39.2	49.9	61.5	72.5	79.4	81.4	79.3	74.6	66.5	53.8	43.4	
10%		WB	34.1	33.8	40.0	48.5	60.2	67.9	71.2	70.0	67.2	58.0	46.5	37.5	
		MCDB	35.9	36.3	45.1	56.1	69.1	75.9	78.7	76.5	72.9	62.7	49.7	39.8	
Mean Daily Temperature Range	5% DB	MDBR	14.8	16.8	17.7	19.6	21.4	21.6	21.3	20.3	21.1	18.2	15.1	13.1	
		MCDBR	13.9	17.2	26.1	30.6	29.2	26.8	24.8	24.2	25.3	24.9	21.1	15.9	
		MCWBR	12.4	13.9	18.1	18.8	15.8	13.4	11.5	12.2	13.4	16.0	16.0	13.6	
	5% WB	MCDBR	13.0	15.9	24.2	27.5	25.7	23.9	21.2	20.8	20.9	21.6	19.2	14.9	
		MCWBR	12.1	13.8	17.9	19.0	15.8	13.8	11.2	11.5	12.3	15.5	15.7	13.6	
Clear-Sky Solar Irradiance	taub	0.281	0.299	0.305	0.351	0.374	0.409	0.407	0.400	0.372	0.344	0.311	0.281		
	taud	2.451	2.373	2.463	2.380	2.387	2.317	2.342	2.355	2.425	2.483	2.524	2.500		
	Ebn at Noon	270	283	297	289	284	273	272	270	270	264	257	259		
	Edh at Noon	23	29	30	36	37	40	38	37	32	26	21	20		
All-Sky Solar Radiation	RadAvg	416	666	1144	1454	1728	1903	1921	1631	1291	781	455	320		
	RadStd	31	89	121	124	152	121	78	67	122	84	59	33		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page